```
from google.colab import files
files.upload()
```

```
Choose Files penguins.csv
penguins.csv(text/csv) - 16735 bytes, last modified: 10/15/2026 - 100% done
Saving penguins.csv to penguins (1).csv
{'penguins (1).csv':
b'species,island,bill_length_mm,bill_depth_mm,flipper_length_mm,body_mass_g,sex,year\nAdelie,Torgersen,39.1,18.7,181.0,3750.0,ma
```

```
import pandas as pd
df = pd.read_csv('penguins.csv')

df.shape
```

```
(344, 8)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   species            344 non-null    object
 1   island             344 non-null    object
 2   bill_length_mm     342 non-null    float64
 3   bill_depth_mm      342 non-null    float64
 4   flipper_length_mm  342 non-null    float64
 5   body_mass_g        342 non-null    float64
 6   sex                333 non-null    object
 7   year               344 non-null    int64
dtypes: float64(4), int64(1), object(3)
memory usage: 21.6+ KB
```

```
df.describe()
```

| | bill_length_mm | bill_depth_mm | flipper_length_mm | body_mass_g | year |
|---|---|---|---|---|---|
| **count** | 342.000000 | 342.000000 | 342.000000 | 342.000000 | 344.000000 |
| **mean** | 43.921930 | 17.151170 | 200.915205 | 4201.754386 | 2008.029070 |
| **std** | 5.459584 | 1.974793 | 14.061714 | 801.954536 | 0.818356 |
| **min** | 32.100000 | 13.100000 | 172.000000 | 2700.000000 | 2007.000000 |
| **25%** | 39.225000 | 15.600000 | 190.000000 | 3550.000000 | 2007.000000 |
| **50%** | 44.450000 | 17.300000 | 197.000000 | 4050.000000 | 2008.000000 |
| **75%** | 48.500000 | 18.700000 | 213.000000 | 4750.000000 | 2009.000000 |
| **max** | 59.600000 | 21.500000 | 231.000000 | 6300.000000 | 2009.000000 |

```
df.head()
```

| | species | island | bill_length_mm | bill_depth_mm | flipper_length_mm | body_mass_g | sex | year |
|---|---|---|---|---|---|---|---|---|
| **0** | Adelie | Torgersen | 39.1 | 18.7 | 181.0 | 3750.0 | male | 2007 |
| **1** | Adelie | Torgersen | 39.5 | 17.4 | 186.0 | 3800.0 | female | 2007 |
| **2** | Adelie | Torgersen | 40.3 | 18.0 | 195.0 | 3250.0 | female | 2007 |
| **3** | Adelie | Torgersen | NaN | NaN | NaN | NaN | NaN | 2007 |
| **4** | Adelie | Torgersen | 36.7 | 19.3 | 193.0 | 3450.0 | female | 2007 |

Next steps: Generate code with df | New interactive sheet

```
df.value_counts()
```

| | | | | | | | | count |
|---|---|---|---|---|---|---|---|---|
| species | island | bill_length_mm | bill_depth_mm | flipper_length_mm | body_mass_g | sex | year | |
| Gentoo | Biscoe | 59.6 | 17.0 | 230.0 | 6050.0 | male | 2007 | 1 |
| Adelie | Biscoe | 34.5 | 18.1 | 187.0 | 2900.0 | female | 2008 | 1 |
| | | 35.0 | 17.9 | 190.0 | 3450.0 | female | 2008 | 1 |
| | | | | 192.0 | 3725.0 | female | 2009 | 1 |
| | | 35.3 | 18.9 | 187.0 | 3800.0 | female | 2007 | 1 |
| | | ... | ... | ... | ... | ... | ... | ... |
| | | 38.1 | 16.5 | 198.0 | 3825.0 | female | 2009 | 1 |
| | | 37.9 | 18.6 | 193.0 | 2925.0 | female | 2009 | 1 |
| | | | | 172.0 | 3150.0 | female | 2007 | 1 |
| | | 37.8 | 20.0 | 190.0 | 4250.0 | male | 2009 | 1 |
| | | | 18.3 | 174.0 | 3400.0 | female | 2007 | 1 |

333 rows × 1 columns

**dtype:** int64

You are helping with exploratory data analysis. My CSV has columns: [species: object], [island: object], [bill_length_mm: float64], [bill_depth_mm: float64], [flipper_length_mm: float64], [body_mass_g: float64], [sex: object]. Write Python using pandas and seaborn to: (1) summarize missing data, (2) compare [flipper_length_mm] across [species], (3) produce one labeled chart. Explain each step in comments.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# --- Step 1: Summarize missing data ---
# We calculate the sum of missing (null) values for each column and display it
print("=== Missing Data Summary ===")
missing_summary = df.isnull().sum()
print(missing_summary)
print("\n")

# --- Step 2: Compare flipper_length_mm across species ---
# We group the data by species and calculate summary statistics for flipper length
print("=== Flipper Length Comparison Across Species ===")
flipper_stats = df.groupby('species')['flipper_length_mm'].describe()
display(flipper_stats)

# --- Step 3: Produce one labeled chart ---
# We create a box plot to visualize the distribution of flipper lengths for each species
plt.figure(figsize=(8, 6))
sns.boxplot(data=df, x='species', y='flipper_length_mm', hue='species', palette='Set2', legend=False)

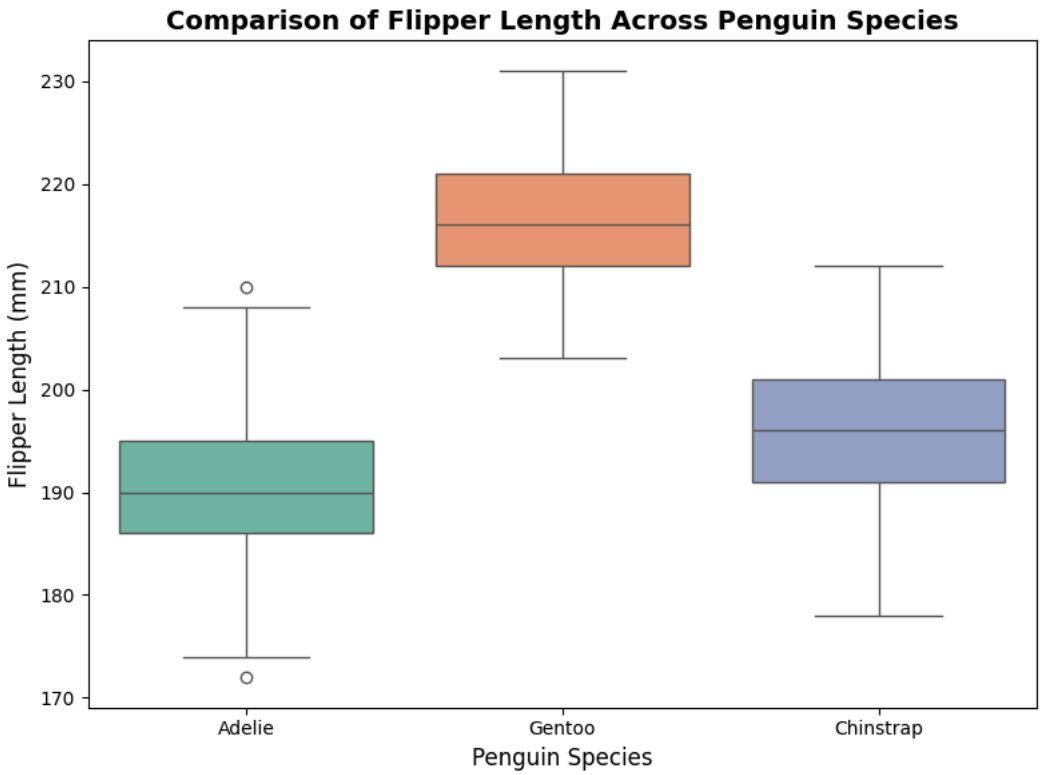
# Add clear labels and a title
plt.title('Comparison of Flipper Length Across Penguin Species', fontsize=14, fontweight='bold')
plt.xlabel('Penguin Species', fontsize=12)
plt.ylabel('Flipper Length (mm)', fontsize=12)

# Adjust layout and display the plot
plt.tight_layout()
plt.show()
```

```
=== Missing Data Summary ===
species               0
island                0
bill_length_mm        2
bill_depth_mm         2
flipper_length_mm     2
body_mass_g           2
sex                  11
year                  0
dtype: int64


=== Flipper Length Comparison Across Species ===
```

| species | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| **Adelie** | 151.0 | 189.953642 | 6.539457 | 172.0 | 186.0 | 190.0 | 195.0 | 210.0 |
| **Chinstrap** | 68.0 | 195.823529 | 7.131894 | 178.0 | 191.0 | 196.0 | 201.0 | 212.0 |
| **Gentoo** | 123.0 | 217.186992 | 6.484976 | 203.0 | 212.0 | 216.0 | 221.0 | 231.0 |




Next steps: Generate code with `flipper_stats` New interactive sheet

For a better prompt, I would write the following:

You are helping with exploratory data analysis on penguins.csv with columns: species (object), island (object), bill_length_mm (float64), bill_depth_mm (float64), flipper_length_mm (float64), body_mass_g (float64), and sex (object). Execute the following Python code using pandas and seaborn: 1. Group the data by both species and sex to calculate multi-level descriptive statistics for flipper_length_mm. 2. Produce a grouped boxplot chart (sns.boxplot with hue="sex") comparing flipper length distributions across species split by sex. Explain each step in comments, run the code, and conclude with a single-sentence entry in an analysis log evaluating whether this multi-variable breakdown improved the insight.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# --- Step 1: Group by both species and sex to calculate descriptive statistics ---
print("=== Flipper Length Descriptive Statistics (Species & Sex) ===")
multi_group_stats = df.groupby(['species', 'sex'])['flipper_length_mm'].describe()
display(multi_group_stats)
print("\n")
```

```
# --- Step 2: Produce a grouped boxplot chart (sns.boxplot with hue="sex") ---
plt.figure(figsize=(10, 6))
# We build the boxplot utilizing 'sex' as the hue to visually segment the distributions
sns.boxplot(
    data=df,
    x='species',
    y='flipper_length_mm',
    hue='sex',
    palette='Set2'
)

# Customize chart labels, legend, and title for clarity
plt.title('Flipper Length Distribution by Species and Sex', fontsize=14, fontweight='bold')
plt.xlabel('Species', fontsize=12)
plt.ylabel('Flipper Length (mm)', fontsize=12)
plt.legend(title='Sex', loc='upper left')
plt.grid(axis='y', linestyle='--', alpha=0.5)

# Display the plot
plt.tight_layout()
plt.show()
print("\n")

# --- Step 3: Conclude with a single-sentence entry in an analysis log ---
print("=== Analysis Log ===")
log_entry = "Analysis Log: Segmenting by sex reveals a consistent, distinct sexual dimorphism across all three species where ma
print(log_entry)
```

```
=== Flipper Length Descriptive Statistics (Species & Sex) ===
```

Next steps: Generate code with multi_group_stats | New interactive sheet

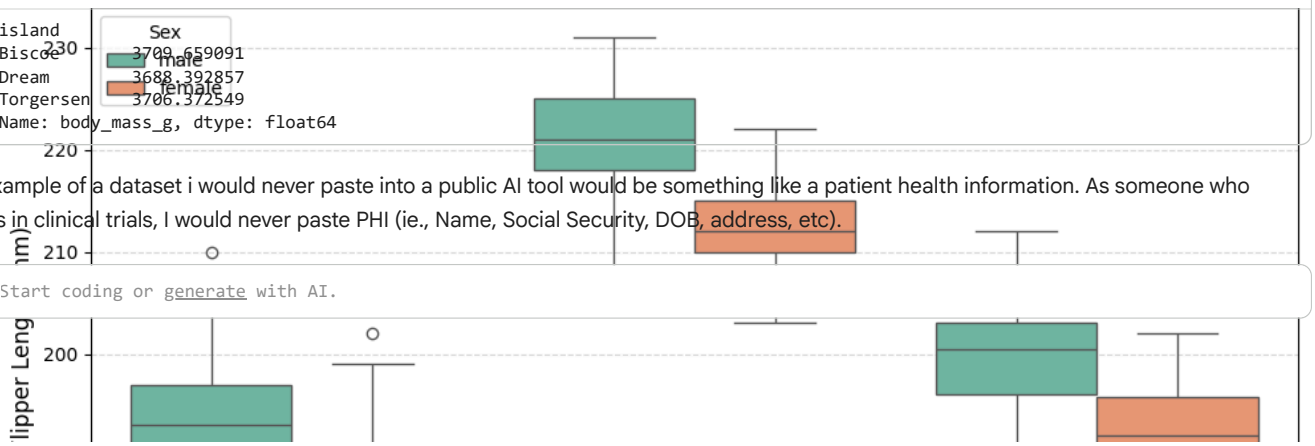
| species | sex | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|---|
| Adelie | female | 73.0 | 187.794521 | 5.595035 | 172.0 | 185.00 | 188.0 | 191.00 | 202.0 |
| | male | 73.0 | 192.410959 | 6.599317 | 178.0 | 189.00 | 193.0 | 197.00 | 210.0 |
| Chinstrap | female | 34.0 | 191.735294 | 5.754096 | 178.0 | 187.25 | 192.0 | 195.75 | 202.0 |
| | male | 34.0 | 199.911765 | 5.976558 | 187.0 | 196.00 | 200.5 | 203.00 | 212.0 |

Something that is not yet discussed or at least something that the AI missed is that the Adelie penguins exhibit significant variations in their body mass depending on where they are. Example, Adelies nesting on Torgersen have a higher average body mass versus those who nest on Biscoe Island.

```
import pandas as pd
df = pd.read_csv('/content/penguins.csv')

adelie_subset = df[df['species'] == 'Adelie']
average_mass = adelie_subset.groupby('island')['body_mass_g'].mean()
print(average_mass)
```

```
island
Biscoe     3709.659091
Dream      3688.392857
Torgersen  3706.372549
Name: body_mass_g, dtype: float64
```




An example of a dataset i would never paste into a public AI tool would be something like a patient health information. As someone who works in clinical trials, I would never paste PHI (ie., Name, Social Security, DOB, address, etc).

```
Start coding or generate with AI.
```